DATA 209
Applied Linear Algebra & Calculus

Instructor Info
Dr. Ron Smith
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Office hours: T/R Noon-2pm, ISC 3378 (no office hours on Jan 22)

Course Overview, Spring 2026
Section 01: MWF 11-11:50am, ISC 2363

This course introduces core mathematical foundations of artificial intelligence and computing. Topics include linear algebra, matrix decompositions, calculus, and differential equations with direct application to machine learning, optimization, and simulation. Students will explore how mathematical constructs such as dot products, eigenvectors, gradients, and transformations underpin AI systems. Concepts are taught with a strong focus on real-world implementation, using computation and visualization to build intuition over symbolic manipulation.

Prerequisites
DATA 101, or CSCI 140 or 141 or equivalent

Topics

Each of the units below will span roughly 2-4 weeks. Adjustments may be made as the course progresses.

Unit 1: Vectors & High-Dimensional Thinking

This unit introduces the idea of data as vectors in a mathematical space. Topics explored include coordinates, dot products, and orthogonality as intuitive geometric ideas, while learning how these relate to practical concepts like similarity and distance in datasets. Applications discussed may include interpreting real-world objects—such as documents and images—as points in high-dimensional space, laying the foundation for many data science workflows.

- Vectors and coordinate systems
- Dot products, the Law of Cosines, and cosine similarity
- Representing data numerically (e.g., text or images)

Unit 2: Matrices as Geometric Transformations

This unit introduces matrices as tools for geometric transformations such as scaling, rotation, and projection. Concepts such as linear independence, dimension, invertibility, eigenvalues, and eigenvectors will be explored from a primarily geometric perspective to gain a deeper intuition of why linear algebra has such a broad variety of applications.

- Matrix-vector multiplication as a geometric transformation
- The identity matrix and concept of a "basis" of a space
- Matrix composition and non-commutativity
- Eigenvectors and eigenvalues
- Dimensionality reduction via projection

Unit 3: Matrix Decomposition and the Structure of Data

In this unit matrix decompositions will be introduced as tools to both reveal the underlying structure of data as well as efficient computational methods.

- Eigendecomposition and diagonalization
- Singular Value Decomposition (SVD)
- Principal Component Analysis (PCA)
- Interpretation of matrix decompositions in data

Unit 4: Calculus, Change, and Optimization

In this unit the fundamental concepts of calculus (derivatives and integrals) are introduced from the perspective of change, accumulation, and optimization. As opposed to more traditional courses in calculus, an emphasis will be placed on numerical calculation rather than symbolic manipulation. Topics related to the inner workings of machine learning and artificial intelligence will be introduced as well.

- What is a derivative? What is an integral? When do we use which?
- Numerical computation of derivatives and integrals
- Loss functions, optimization, and convergence

Unit 5: Systems, Dynamics, and Modeling with Differential Equations

In this unit we will combine what we have learned about linear algebra and calculus to understand how systems that change over time can be modeled mathematically, as well as continuing our discussion on how machine learning and AI models are optimized. Emphasis will be placed on attaining a high-level understanding of how these methods are applied rather than symbolic manipulation or being able to code your own versions from scratch.

- Systems of differential equations
- Solving linear systems numerically
- Gradients and gradient descent
- Local vs. global minima/maxima
- Efficient optimization methods (e.g., Stochastic Gradient Descent (SGD) and related variants)

Grading

Checkpoints & Participation: 20%
Short checkpoint assignments will be given on an approximately bi-weekly basis. These checkpoints will have unlimited attempts (up to the due date) and primarily graded on a completion basis (unless you clearly made no effort) with the goal of ensuring that are you are sufficiently tracking key concepts. **Use these as an opportunity to ensure that you are grasping the material before the real assignments come**. Some checkpoints will be completed as in-class activities. This, together with your overall participation and level of engagement will constitute 20% of your grade.

Assignments: 50%
More comprehensive assignments will be given, also on an approximately bi-weekly basis. These will be graded on a 10-point scale and will test your ability to apply the methods we have learned to practical datasets.

Final Project 30%
For your final project, you will work in small groups (2-3 students) to optimize a small machine learning model, or design a visual demonstration related to optimization. We will discuss potential projects in more detail in class. You will give a brief in-class presentation and submit a written report documenting your methodology, and reflect upon challenges you encountered and what you learned. The final written report will be due at the time assigned for final exams by the registrar.

Final Grades
Final letter grades are assigned based on simple 10-point intervals, 90-100 = A, 80-89=B, etc. Plus/minus grades will only be assigned in marginal cases and will only ever benefit you. For example, if your final grade was computed to be an 88 or 89, but you have been consistently participating in class, putting forth notable effort on your work and regularly coming to office hours, this would likely be considered sufficient to increase your final grade to a B+ or A-.

Attendance
If you need to miss class—due to illness, family responsibilities, athletic or academic travel, or other personal reasons—please let me know in advance whenever possible. Occasional absences are expected and will not affect your grade if you remain engaged overall.

Consistent engagement is key. If you are frequently absent without prior communication, your ability to participate meaningfully will naturally be limited, which will reduce the participation portion of your grade. At the end of the semester, I can only evaluate the engagement you've demonstrated throughout the course.

If you anticipate ongoing challenges (medical, family, or otherwise), I encourage you to speak with me early so that we can work together on reasonable solutions. Students with documented accommodations should also coordinate with the university's disability services

office to ensure full support.

Lateness
Late assignments are not accepted will automatically be assigned a zero. If there are truly extenuating circumstances this should be addressed with me as soon as is reasonably possible. Requests for extensions made after the fact, or shortly before an assignment is due, will not be considered.

Academic Integrity & the Honor Code

Unless otherwise stated, you are free to use any and all online resources available to you. If an assignment states that certain tools or resources are not permitted, you must abide by this. Likewise, and unless otherwise stated, you may work with others in the class as long as your collaboration is genuine and does not involve simply comparing answers or sharing code.

In all cases, you must report what tools and resources were used, and if you worked with others, with whom and specifically on what.

When in doubt, ask me for clarification ahead of time. Any violations of this policy or perceived dishonesty will not be tolerated and may result in your being reported for an Honor Code Violation. More details on the W&M Honor Code and potential consequences for violations can be found here:
https://www.wm.edu/offices/communityvalues/sarp/honorcodeandcouncils/

In general, make sure you are acting in good faith and focused on learning at all times, and we should not have any problems.

Important dates

Add/drop deadline: January 30
Withdrawal period begins: January 30
Spring Break: March 7-15
Withdrawal deadline: March 23

Accommodations
It is the policy of William & Mary to accommodate students with disabilities and qualifying diagnosed conditions in accordance with federal and state laws. Any student who feels s/he may need an accommodation based on the impact of a learning, psychiatric, physical or chronic health diagnosis should be referred to Student Accessibility Services staff at 757-221-2509 or at sas@wm.edu. SAS staff will work with you to determine if accommodations are warranted, and if so, to help you obtain an official letter of accommodation.

Mental and Physical Well-Being

William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386
- For additional support or resources & questions, Contact the Dean of Students at 757-221-2510 or deanofstudents@wm.edu

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Student Success
Student Success supports students' personal growth, development, engagement, and belonging by providing holistic guidance as they navigate their own W&M journey. It encompasses the offices of Academic Wellbeing, Care Support Services, and Student Accessibility Services.

- For academic support such as tutoring, time management, study skills, and academic coaching, please contact Academic Wellbeing at wm.edu/academicwellbeing (academicwellbeing@wm.edu)
- For concerns about the wellbeing of a member of the William & Mary community or to seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu)
- For accommodation needs or questions, please contact Student Accessibility Services at wm.edu/sas (sas@wm.edu)